



Influence of Humidity on the Tensile Strength of 3D Printed PLA Filament

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Abstract. Polylactic acid (PLA) filament is sensitive to humidity due to its hygroscopic properties, resulting in poor printing quality. Many studies focus on the effect of process parameters on the 3D printed part; however, the influence of humidity on the printing quality is less explored. Moisture can influence the mechanical properties of 3D printed parts due to the water's existence, altering the polymer chain's bonding, resulting in a lower mechanical strength. This study investigates the effect of humidity on the tensile strength of 3D printed PLA specimens exposed to various humidity conditions. Three conditional settings were established; a new PLA roll acts as the reference, used PLA roll stored in the vacuum bag with desiccant, and used PLA roll stored in an open environment, exposed with the humidifier for 24–150 h. As a result, the moisture decreases the tensile strength and increases the plastic region. The presence of moisture causes a plasticizing effect of the polymer, which increases its deformability while lowering its rigidity as the moisture changes the polymer's molecular structure. Humidity expands the interlayer gap, promoting enhancement in the fracture strain due to increased macromolecules deformation space. However, the used PLA filament stored in a vacuum bag with some desiccants shows an equivalent tensile strength with the reference specimen. To conclude, humidity influences the tensile strength and should be controlled for a good printing quality of PLA parts. Therefore, it is recommended that all 3d printer users seal the used PLA filament in a vacuum bag with some dehumidifying agents as an alternative to the drying cabinet.

Keywords: Humidity · Tensile strength · PLA

1 Introduction

Additive manufacturing technologies play an essential role in Industry 4.0. One of the most prevalent processes is Fused Deposition Modelling (FDM) due to its versatility and low cost [1]. Various factors influence the material properties of filaments. One of the significant factors affecting the characteristics of filaments is humidity. For example, if the polylactic acid (PLA) filament is exposed to high humidity for a long time, the filament has decreased tensile strength at the time of printing and generates bubbles that occur on the surface of the output [2]. In addition, PLA is sensitive to humidity, and exposure to high humidity causes changes in its properties. It has been reported that as

humidity increases, average molecular weight, elongation percentage, and degradation rate decrease [3].

Polyesters are widely recognized for absorbing humidity, and the presence of moisture causes severe degradation events. The degradation is mainly caused by the hydrolysis of macromolecular chains, which significantly reduces molecular weight [4]. According to Lesaffre et al. [5], PLA is sensitive to such environmental restrictions. Filaments made off PLA happened to have hygroscopic characteristics that appear to absorb moisture from the air, contributing to inconsistent and poor printing quality. Even though humidity is one of the significant factors affecting the mechanical properties of PLA printed specimens, a study on this is still lacking. It is expected that with the increase of exposure time with a humid condition, the moisture content in the PLA filament increases and simultaneously decreases the tensile strength of the printed parts. Therefore, the objective of this study is to investigate the effects of humidity on the tensile strength of the PLA specimens exposed to various humidity conditions.

2 Materials and Method

In this study, the commercial Cixi Lanbo PLA filament of a blue colour, manufactured by Cixi Lanbo Printer Consumable CO., LTD. with a diameter of 1.75 mm, was used as the printing material. Three different conditions of the PLA filament were established, as shown in Table 1. The first group was the new PLA filament roll which acts as the reference, and the second group was the used PLA filament, which was stored in the vacuum bag with desiccant. The third group was the used PLA filament exposed to humidity for 24–150 h. The number of replications for each conditional setting was set to three to reduce random errors.

Table 1. The type of PLA filaments conditions and number of specimens.

Condition of PLA filaments		Number of specimens
a. New PLA filament roll, brand new which acts as the reference		3
b. Used PLA filament roll stored in the vacuum bag, with desiccant		3
c. Used PLA filament roll stored in open environment, exposed to a humidifier for a variant of:	24 h	3
	48 h	3
	72 h	3
	96 h	3
	120 h	3
	150 h	3

The first group was the reference in this study, where the new PLA filament was unsealed and directly used for printing. However, for the second group, the used filament was stored with 50 g of desiccants in a vacuum bag. Finally, the third group was exposed

to various humidity conditions using the house humidifier, continuously switched on according to the set time.

The PLA samples were manufactured using FlashForge’s Adventurer3 3D printer, a low-cost desktop printer with a 0.4 mm nozzle size. Table 2 depicts the setting of process parameters used in this study. Firstly, the printing temperature was set to 210 °C, following the recommended printing temperature of PLA filaments, which is in the range of 180 °C to 210 °C [6]. According to Vallergera et al. [2], a better mechanical strength is achieved at a higher temperature. As for the rest of the process parameters, a default setting of the PLA was applied.

Table 2. Setting of process parameters for 3D printing.

Parameter	Setting
Printing temperature (°C)	210
Bed temperature (°C)	60
Printing speed (mm/s)	60
Filling percentage (%)	100
Material	Polylactic acid (PLA)
Layer thickness (mm)	0.18 (default)

The 3D printed specimens’ geometry was modelled using CATIA V5R21 software, complying with ASTM D638 Type IV standard, as shown in Fig. 1. FlashPrint software is adapted in setting the required parameters and finally generating the G-code. Shimadzu AGS-X was used for the tensile test with a 20 kN load. Tensile testing was performed with a testing speed of 5 mm/min.

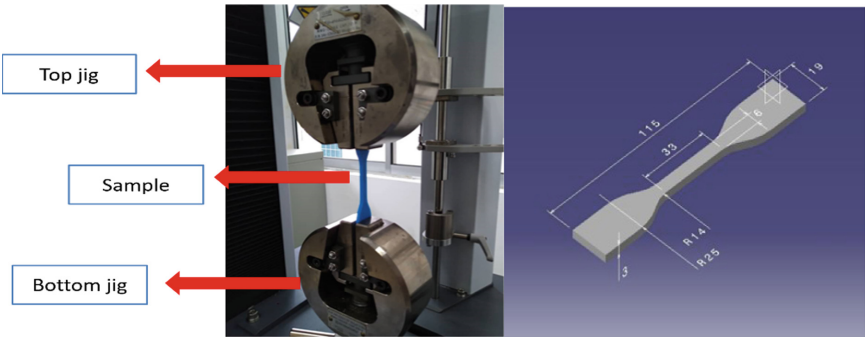


Fig. 1. Tensile test set-up (left), and the ASTM D638 Type IV specimen (right).

3 Results and Discussion

Table 3 summarizes the maximum force and maximum stress for all conditions. According to Fig. 2, the reference sample exhibits the highest maximum force, and 150 h of exposure to humidity was the lowest. The higher the maximum force, the higher the tensile strength. The data shows a decreasing trend. Therefore, the sample printed using the new PLA filament (reference) has the highest tensile strength, while the sample printed using the PLA filament humidified for 150 h has the lowest tensile strength. If PLA filaments are exposed to humidity, the tensile strength of the 3D printed object is decreased.

Table 3. Maximum force and stress for all conditions

Humidity condition	Max force (Avg.) (N)	Max stress (Avg.) (N/mm ²)
New PLA filament roll, which acts as the reference	939.76	16.49
Used PLA filament roll stored in the vacuum bag, with an addition of desiccant	938.72	16.47
Humidified for 24 h	860.29	15.09
Humidified for 48 h	855.63	15.01
Humidified for 72 h	852.78	14.96
Humidified for 96 h	850.48	14.92
Humidified for 120 h	848.69	14.88
Humidified for 150 h	836.47	14.68

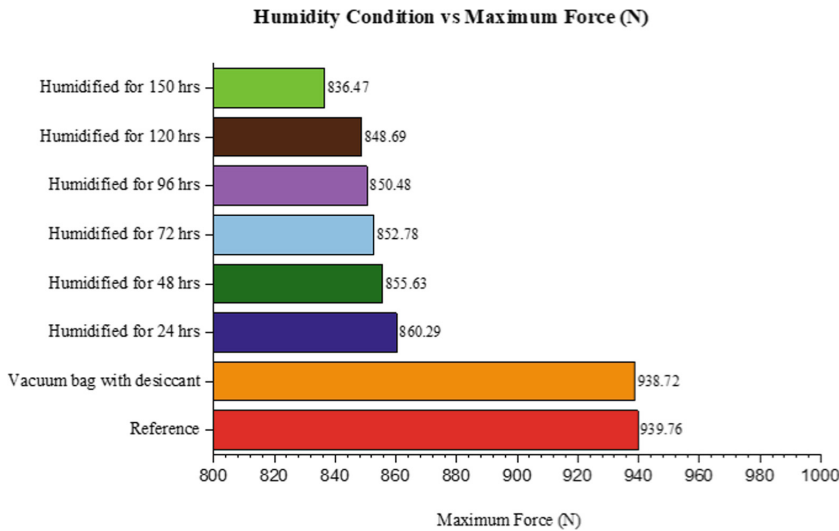


Fig. 2. Maximum force plot for all conditions.

The effect of moisture absorption on stress-strain behavior is seen in Fig. 3. The reference's ultimate tensile strength (UTS) was the highest, while PLA filament humidified for 150 h was the lowest. The strain value of the necking region (strain value from UTS to fracture) varies depending on the humidity level; however, the lengths were uneven. The fracture strain for all humidified samples was extended to the other two conditions with no moisture exposure. Humidity causes the PLA to plasticize, which increases its deformability while lowering its rigidity as the moisture changes the polymer's molecular structure. Humidity expands the interlayer gap, promoting enhancement in the fracture strain due to increased macromolecules deformation space. The increase in water content corresponding with the decrease in stress and the increase in strain may result from the hydrogen bonding between polymer chains. The water molecules can separate into two parts: free water that can freely transfer from holes to polymer chains, and the other is bound to water that can interact with functional groups of material [7]. As a result, the plasticizing impact of moisture resulted in an increase in elongation at break (fracture strain) and a decrease in tensile strength.

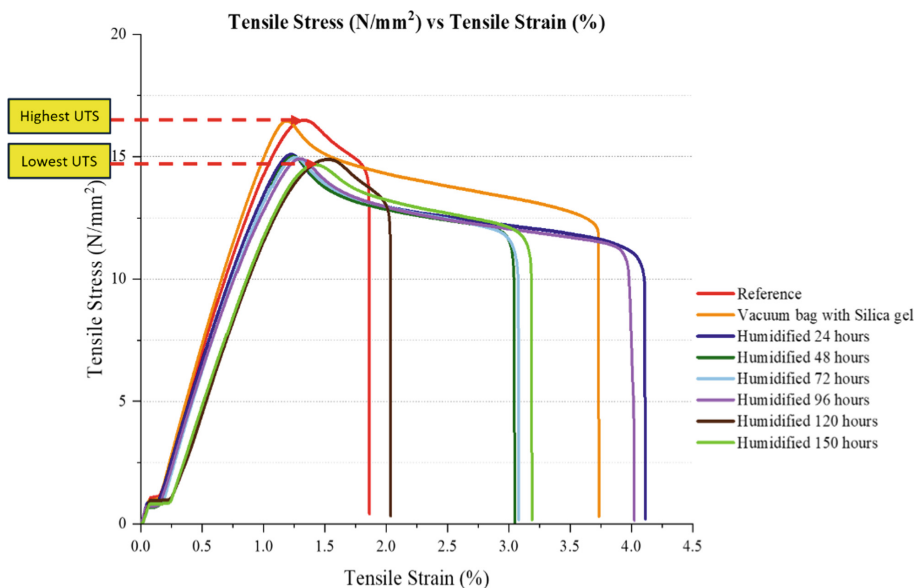


Fig. 3. Stress and strain plot.

4 Conclusion

To conclude, humidity influences the tensile strength of PLA parts. The stress and strain plot demonstrated an increase in the plastic region of all specimens exposed to various humidity conditions and decreased tensile strength. The presence of moisture strongly modifies the mechanical properties. The presence of moisture causes the polymer's

plasticizing effect, which increases the elongation at break. However, the vacuum bag with the addition of desiccant is sufficient to control the humidity at a more reasonable cost than the drying rack. The average tensile strength of these samples is equivalent to the reference specimens. Therefore, it is sufficient for FDM users to store the used PLA roll in a vacuum bag, with some dehumidifying agent to avoid moisture absorption, in substitution of the drying rack, which is more expensive.

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